

#1

$$\frac{dm_{c,sys}}{dt} = -0.116 m_{c,sys} \quad \text{mass of caffeine in the body}$$

$$m(t) = m_0 e^{-kt}, \quad k = 0.116 h^{-1}$$

⇒ Caffeine drop from 200 mg to 100 mg

$$\frac{100}{200} = e^{-0.116t}$$

$$t = \frac{\ln(2)}{0.116} \Rightarrow t = 5.97 h$$

• it takes about 6 hours to drop to 100 mg

time	hours since 6 A.M.	$m_{total}(t)$
6 AM	0	200
8 AM	2	$200 \times e^{(-0.116)(2)} = 159 \text{ mg}$
12 PM	6	$200 \times e^{(-0.116)(6)} = 100 \text{ mg}$
2 PM	8	$200 \times e^{(-0.116)(8)} = 77 \text{ mg} + 200 = 277 \text{ mg}$
6 PM	12	176 mg
11 PM	17	98 mg

* Without the 2 PM mug $\rightarrow$ by 11 PM ✓
Caffeine = 28 mg

* With the 2 PM mug by 11 PM ✓
Caffeine = 98 mg

15
✓

$\Rightarrow$ so cutting out the second mug would leave about $\frac{1}{4}$
as much caffeine by bedtime

good.

#2

$$\frac{dm}{dt} = \text{rate in} - \text{rate out} - \text{rate degraded} \quad \checkmark$$

$$\Rightarrow \frac{dm}{dt} = 62000 - 57000 - 2700(1 - e^{-0.3t})$$

$$= 5000 - 2700 e^{-0.3t}$$

$$= 2300 + 2700 e^{-0.3t} \quad \checkmark$$

$$\Rightarrow m(t) = \int_0^t (2300 + 2700 e^{-0.3t}) dt$$

$$= 2300t + \frac{2700}{0.3} (1 - e^{-0.3t}) \quad \checkmark$$

$$= 2300t + 9000(1 - e^{-0.3t}) \quad \checkmark$$

$$\Rightarrow m(8) = 2300(8) + 9000(1 - e^{(-0.3)(8)})$$

$$= 18,400 + 9000(1 - e^{-2.4}) \quad \checkmark$$

$$\Rightarrow e^{-2.4} = 0.0907$$

$$1 - e^{-2.4} = 0.9093$$

$$9000(0.9093) = 8184$$

15

$$\Rightarrow m(8) = 18,400 + 8184$$

$$= 26,584 \text{ molecules} \quad \checkmark \quad \checkmark$$

* the immunotoxin does not accumulate fast, so we should not invest invest

#3

$$D = 1 \text{ m}, \quad A = \pi (0.5)^2 = 0.785398 \text{ m}^2$$

$$\rho = 1000 \text{ kg/m}^3, \quad \text{feed solution} = 40 \text{ kg/min containing } 10\% \text{ X}$$

• before the leak started $h_0 \rightarrow 1 \text{ m}$

$$m = \rho A h \rightarrow \text{No leak}$$

$$\rho A \frac{dh}{dt} = \dot{m}_{\text{in}} = 40 \rightarrow \frac{dh}{dt} = \frac{40}{\rho A}$$

$$t(1) = \frac{\rho A h}{40} = \frac{1000 \times 0.785398 \times 1}{40} = 19.635 \text{ min}$$

after the leak starts
mass balance

$$\rho A \frac{dh}{dt} = 40 - 4\sqrt{h-1} \quad (h \geq 1)$$

$$dt = \int_{h_0}^0 \rho A \frac{2y}{40-4y} dy$$

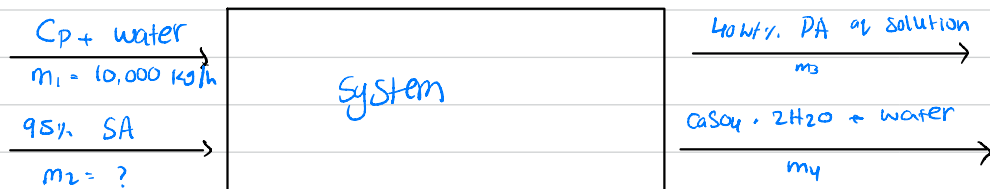
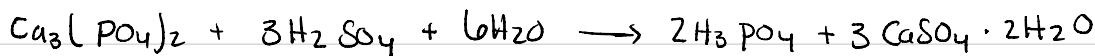
$$\Delta t = 40 - t_1 = \rho A \left[-\frac{y_{40}}{2} - 5 \ln \left(1 - \frac{y_{40}}{10} \right) \right]$$

Put this equation into excel

there are missing steps.

15/20

#4



$$\text{Ca}_3(\text{PO}_4)_2 = 310.18 \text{ kg/kmol}$$

$$\text{H}_2\text{SO}_4 = 98.08 \text{ kg/kmol}$$

$$\text{H}_2\text{O} = 18.02 \text{ kg/kmol}$$

$$\text{H}_3\text{PO}_4 = 98.00 \text{ kg/kmol}$$

$$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} = 172.17 \text{ kg/kmol}$$

* moles of CP Fed:

$$\dot{n}_{\text{CP}} = \frac{10000}{310.174} = 32.240 \text{ kmol/hr} \quad \checkmark$$

$$\bullet \dot{n}_{\text{SA}} = 3\dot{n}_{\text{CP}} = (3)(32.240) = 96.720 \text{ kmol/h} \quad \checkmark$$

$$\bullet \dot{n}_{\text{PA}} = 2\dot{n}_{\text{CP}} = 64.480 \text{ kmol/hr} \quad \checkmark$$

$$\bullet \dot{n}_{\text{gypsum}} = 3\dot{n}_{\text{CP}} = 96.72 \text{ kmol/hr} \quad \checkmark$$

$$\bullet \dot{n}_{\text{H}_2\text{O to gypsum}} = 2\dot{n}_{\text{gypsum}} = 193.440 \text{ kmol/hr} \quad \checkmark$$

$$\bullet \text{Pure SA needed: } (96.72)(98.077) = 9486.00 \text{ kg/hr} \quad \checkmark$$

$$\bullet \text{PA made: } (64.480)(97.994) = 6318.63 \text{ kg/hr} \quad \checkmark$$

$$\bullet \text{Gypsum made: } (96.72)(172.169) = 16652.17 \text{ kg/hr} \quad \checkmark$$

Stream 2 total (95 wt% SA)

$$\dot{m}_2 = \frac{9480.0}{0.95} = 9985.26 \text{ kg/hr}$$

$$\text{SA} = 9480.0, \quad \text{water} = 499.26$$

⇒ Stream 3 (40 wt% PA)

$$\dot{m}_3 = \frac{6318.63}{0.40} = 15796.57 \text{ kg/hr}$$

$$\text{Water} = 0.60 \dot{m}_3 = 9477.94$$

⇒ Stream 4 (gypsum)

$$\dot{m}_4 = 16052.17 \text{ kg/hr}$$

$$\text{CS part} = 13167.37, \quad \text{bound water} = 3484.80$$

⇒ water in Stream 1

$$W_1 + 499.26 = 9477.94 + 3488.80$$

$$W_1 = 12463.48 \text{ kg/hr}$$

⇒ Check over all mass balance

$$10000 + 12463.48 \text{ S1} + 9985.26 \text{ S2} = 15796.57 \text{ S3} + 16052.17 \text{ S4} \\ = 32448.75 \text{ kg/hr}$$

mass in = mass out

b) $x(0) = 0.20$, $M = 10,000$ kg, $\dot{m} = \dot{m}_3 = 15796.57$ kg/hr

$$\frac{dx}{dt} = \frac{\dot{m}}{M} (x_{in} - x)$$

$$\Rightarrow x(t) = x_{in} + (x_0 - x_{in}) e^{-(\dot{m}/M)t}$$

at $t = 1$ hr:

$$x(1) = 0.40 + (0.20 - 0.40) e^{-1 \cdot 15796.57}$$

$$= 0.3588 \rightarrow 35.9 \text{ wt\% PA}$$

$$\frac{20}{20}$$

#5a

- w = mass fraction of ethanol in tank (wt.%)
- m = total mass of solution in tank

$$\frac{d(wm)}{dt} = \dot{m}_{in} w_{in} - \dot{m}_{out} w$$

for pure water : $w_{in} = 0$

$$\Rightarrow \dot{m}_{in} - \dot{m}_{out} = F_m$$

$$m \frac{dw}{dt} = -F_m w \Rightarrow \frac{dw}{dt} = - \frac{F_m}{m} w$$

$$w(t) = w_0 e^{-\frac{F_m}{m} t}$$

from the website

$$\rho_{95} = 0.804 \text{ kg/L}$$

$$m = (0.804)(1000) = 804 \text{ kg}$$

Flow 100 L/min $\rightarrow$ mass flow rate :

$$F_m = (100)(0.804) = 80.4 \text{ kg/min}$$

$$\frac{w_0}{w} = \frac{5}{95} = 0.05263$$

$$t = \frac{m}{F_m} \ln \frac{w_0}{w} = \frac{804}{80.4} \ln \left(\frac{95}{5} \right)$$

$$t = 10 \ln(19) = 29.4 \text{ min}$$

$$23.72 \text{ min}$$

$$\dot{m} = \rho (w_{eth}) V =$$

$$\rightarrow \text{Water} \rightarrow \rho_w \cdot \dot{V}_1 = 99.823 \frac{\text{kg}}{\text{min}}$$

6/10

b) ethanol mass in tank: $M_e = \rho V w$

with pure water in, the ethanol balance is

$$\frac{d(\rho V w)}{dt} = -\rho w Q \rightarrow \frac{d(\rho w)}{dt} = -\frac{Q}{V} \rho w \quad \checkmark$$

let $y = \rho w$

then $\dot{y} = -\frac{Q}{V} y \rightarrow y(t) = y_0 e^{-(Q/V)t} \quad \checkmark$

$$\Rightarrow t = \frac{V}{Q} \ln \left(\frac{y_0}{y_f} \right) = \frac{V}{Q} \ln \left(\frac{\rho(w_0) w_0}{\rho(w_f) w_f} \right)$$

from the website

$$\rho(95\%) = 0.80424 \text{ kg/L}$$

$$\rho(5\%) = 0.98938 \text{ kg/L}$$

$$y_0 = \rho(0.95) 0.95 = 0.80424 \times 0.95 = 0.76403$$

$$y_f = \rho(0.05) 0.05 = 0.98938 \times 0.05 = 0.04947$$

$$t = \frac{1000}{100} \ln \left(\frac{0.76403}{0.04947} \right)$$

$$= 10 \ln(15.443)$$

$$= 27.4 \text{ mins} \quad \checkmark \quad 20/20$$

It's faster in b because the denser leaving late carries more ethanol per liter than if density had stayed at the initial value